

Artificial intelligence in Africa: a bibliometric analysis from 2013 to 2022

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Abstract

This study employs bibliometric analysis to investigate the evolving research landscape of Artificial intelligence (AI) within Africa, focusing on the years 2013 to 2022. The central objective is to discern and analyze AI studies conducted in Africa, using a dataset compiled from research papers within the Scopus database. By conducting a comprehensive analysis, this research uncovers crucial insights, including primary authors, influential journals and publishers, nations with the highest research productivity, noteworthy funding sources, influential organizations, and prevalent research domains. Additionally, the study examines year-by-year growth trends and authorship patterns. Employing the VOSviewer software, it creates visual representations that illustrate the dynamic evolution of AI research within the African context. Notably, the analysis of 1646 publications reveals a significant increase in publications over the last decade, with South Africa emerging as a global leader in AI development, and the IEEE, Elsevier, and Springer as prominent publishers. The study also highlights the leading institutions, with the University of the Witwatersrand, University of Johannesburg, University of KwaZulu-Natal, University of Cape Town, and University of Pretoria at the forefront of AI research in Africa. The National Research Foundation is identified as the primary funder supporting AI research across the continent. In conclusion, this research aims to provide a comprehensive understanding of AI's role in addressing African challenges, fostering innovation, and contributing to the continent's technological advancement, shedding light on prevalent research areas and significant funding sources in the process.

Keywords Artificial intelligence · Machine learning · Deep learning · Healthcare · Bibliometric analysis

1 Introduction

Artificial intelligence (AI) is increasingly receiving attention and interest from academia, industry, governmental agencies and the society at large. AI has greatly advanced human capabilities in undertaking complex activities from various domains such as healthcare, infrastructure and data ecosystem, digital economy, environmental conservation, and agriculture. Advancement in AI presents an opportunity to evolve better solutions towards addressing prevalent societal challenges and deliver long-lasting positive impacts [1]. As a tool that can be utilized to improve and advance every aspect of our life, AI has become ingrained in the lives of citizens of the twenty-first century [2]. It does this by using a set of mathematical algorithms to identify a familiar pattern from the given input data and form a set of rules that can be used to “teach” the machine to identify, classify and extract information [3].

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AI tries to replicate and augment human intelligence through technology. Some of the research fields in AI include machine learning, computer vision, natural language processing, robotics, expert systems, reinforcement learning, cognitive computing, neural networks, knowledge representation and reasoning. The science of AI merges numerous academic disciplines and covers a wide range of methodologies [4, 5]. According to [6], Computers may learn from data without explicit programming thanks to machine learning (ML). ML systems employ algorithms to extract features from a given collection of data. The benefit of ML is that it increasingly changes as it encounters more input data [7].

Through supervised learning, which entails providing them with labelled data, machines learn. By giving our computer access to a lot of data and instructing it on how to examine it, we are teaching it in this process. Unsupervised learning methods, as opposed to supervised learning, are used by machines to analyze unlabeled data. The program independently comes to a conclusion after looking for patterns in the data. It's important to keep in mind that the computers make assumptions and that the dataset used in this instance is not labelled. Reinforcement learning is a type of machine learning model that depends on feedback. In this procedure, data is provided to the computer and it is asked to predict what the data will be. If the machine incorrectly interprets the incoming data and makes a decision, it is informed of its faults.

Deep learning is a method used in artificial intelligence (AI) that teaches computers to analyze data in a way that mimics the human brain. In order to produce accurate analyses and projections, deep learning models can recognize complex patterns in photos, text, audio, and other sorts of data. It is the theory that computers can replicate the processes used by the human brain to reason, analyze, and learn. As part of an AI's cognitive process, a neural network is employed in deep learning. Deep learning needs a very powerful processing machine and a lot of training data.

With the use of Deep learning technique, unstructured data may be given structure, and computers may be taught to automatically classify data. Deep learning systems, which may be classed into supervised and unsupervised learning, aspire to become the AI brain capable of managing high volume and high dimensional data through feature extraction [8].

A branch of artificial intelligence called "natural language processing" (NLP) aims to get computers as close to a human-level understanding of language as possible by enabling them to comprehend and process human languages. According to Hirschberg and Manning [9], NLP focuses on using computational methods to learn, comprehend, and synthesize human language content. NLP, or natural language processing, makes it possible for humans and robots to communicate.

African countries have a great possibility to use AI to boost their competitiveness. In a number of industries, including agriculture, transportation, finance, energy, water, e-commerce, education, data security, language translation, and remote health, AI can help Africa overcome its economic issues. To analyze the state of artificial intelligence research in Africa at the present time, this study used bibliometric analysis and visualization. By doing this, future researchers will be able to understand the current state of AI research in Africa.

This study explores the history and trends of AI in Africa by addressing the following research questions:

1. What are the distribution patterns of articles in the AI area in Africa?
2. Which are the most active academic institutions and countries in the AI area in Africa?
3. Who are the highly productive authors of AI in Africa?

The remainder of this paper is structured as follows. Section 2 describes the research methodology, while Sect. 3 presents the findings. Section 4 provides the discussions of the study, while Sect. 5 provides the conclusions and recommendations of the study.

2 Methods

2.1 Data source

Two members of the research team searched the Scopus core collection database for articles published between January 2013 and December 2022 in order to conduct a bibliometric analysis. On April 2, 2023, researchers looked for articles published on AI. ("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Natural Language Processing") AND Africa is the search term utilized in this investigation. 1646 academic publications in total were retrieved from the Scopus database.

2.2 Inclusion and exclusion criteria

Papers published in the Scopus database from January 2013 to December 2022 were used in the bibliometric analysis. Furthermore, final articles which were written in English were involved in the bibliometric analysis. The exclusion criteria were as follows: (i) documents which were written in non-English language and (ii) documents which were published in other databases as well as papers which were not published between January 2013 and December 2022.

2.3 Data analysis

Keyword co-occurrence and country co-authorship in the bibliography were evaluated using the VOSviewer (V.1.6.17) software application for building and displaying bibliometric networks. Using Microsoft Excel, the researchers carried out descriptive analyses to assess the qualities and kinds of articles obtained. Publishers, keywords, categories, sources, funders, author information, author nations, and author connections were among the analyzed items.

3 Results

3.1 Publications output

The results in Table 1 shows that a total of 1646 eligible publications were selected, of which 952 (57.84%) were articles, 466 (28.31%) were conference papers, 70 (4.25%) were reviews, 65 (3.95%) were Conference Reviews, 38 (2.31%) were Book Chapters, 23 (1.40%) were Notes, and 32 (1.94%) were other types of publications. This shows that more than half of the publications (57.84%) were journal articles.

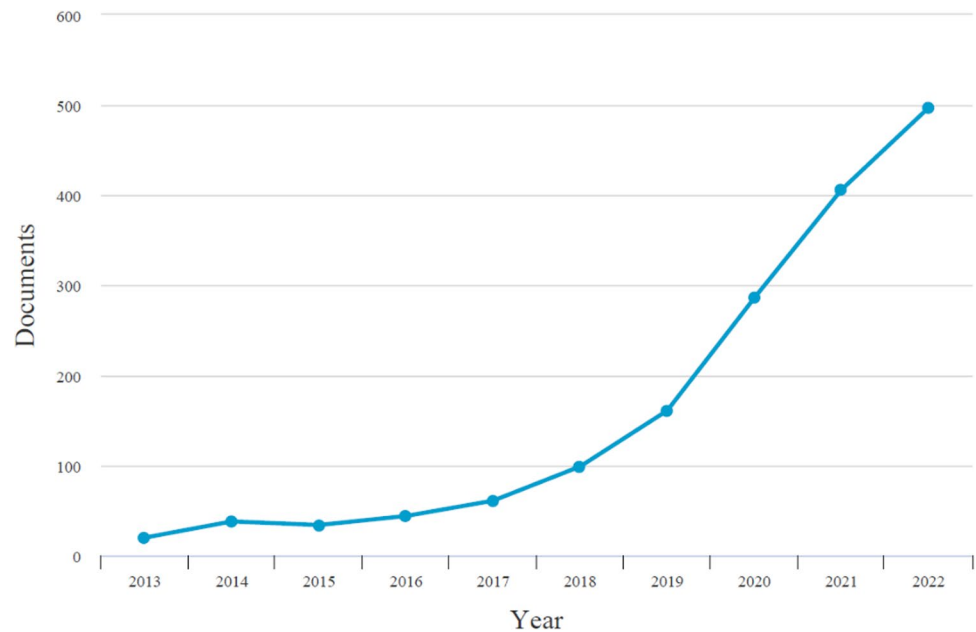
3.2 Growth trend of publications

There were 1646 AI documents published in Scopus database, according to the data. The number of publications on AI in Africa significantly increased between 2013 and 2014, whereas it significantly decreased between 2014 and 2015, as seen in Fig. 1. After 2015, there was a shift in which the quantity of publications on AI in Africa significantly increased. There are no clear indications why the publication trend has decreased between 2014 and 2015. However, the increase in number of publications from 2015 to 2022 can partly be caused by the fact that academics are now publishing in this area, and that their work is available online. It could be difficult to gauge the scope of ongoing investigations when there is no reporting on that research. In actuality, not all journals published by universities and other organizations are accessible online. The Scopus databases can only be used to gather online-only sources.

There was a significant global geographic dispersion for the articles published in this subject between 2013 and 2022, and there was a quick increase in their number (Table 2). In comparison to 2013, there were 477 more articles published

Table 1 Document counts by type

Document type	Counts	Percentage
Article	952	57.84
Conference Paper	466	28.31
Review	70	4.25
Conference Review	65	3.95
Book Chapter	38	2.31
Note	23	1.40
Editorial	14	0.85
Data Paper	7	0.43
Book	4	0.24
Short Survey	3	0.18
Erratum	2	0.12
Letter	2	0.12
Total	1646	100

Fig. 1 Scholarly works over time

in 2022 than there were in 2013. It can also be evidenced that 30.19% of the total publications were published in 2022, 24.67% were published in 2021, 17.38% were published in 2020 and 9.78% were published in 2019 during the study period. Publications published in other years were less than 20% in total publications. This shows that a greater number of AI literature have been published between 2019 and 2022.

3.3 The distribution of AI study published sources

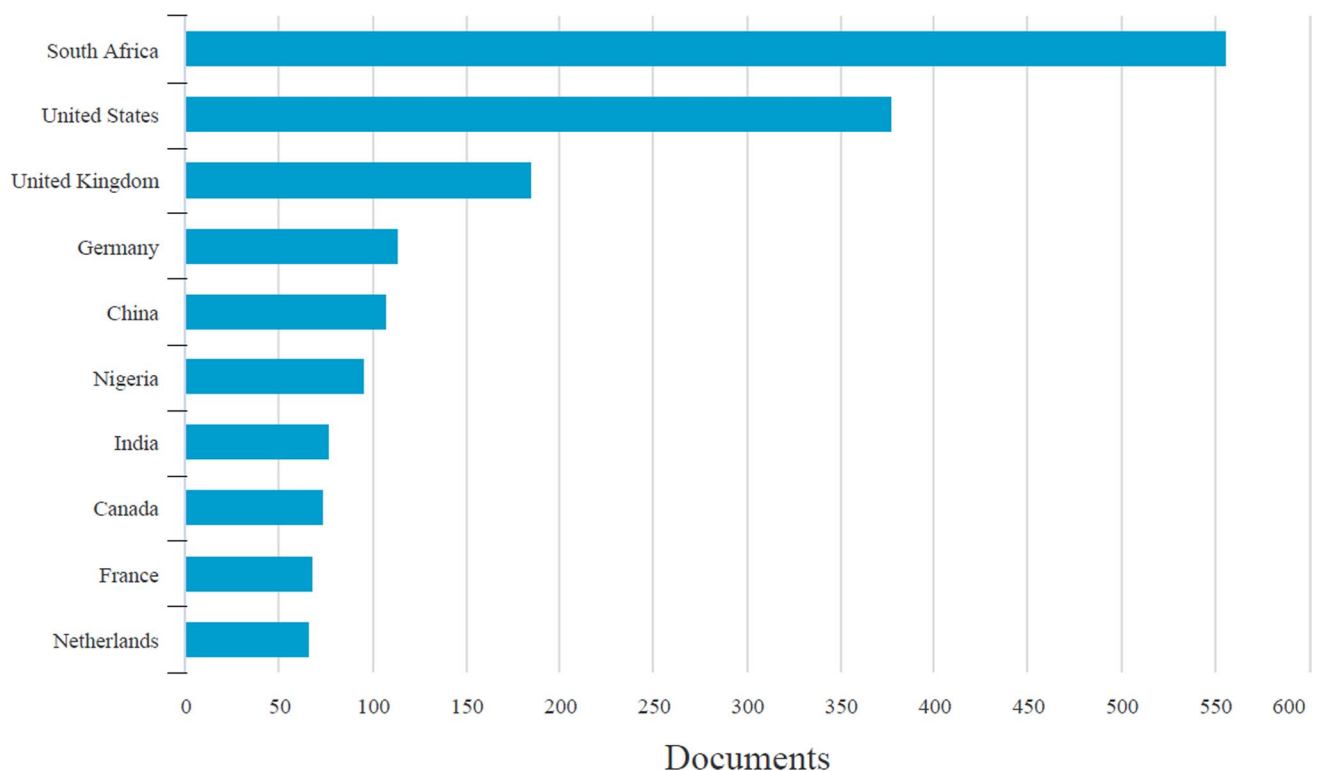
Remote Sensing, which has 53 papers, is the source with the greatest significant number of papers. The second is international geoscience and remote sensing symposium (igarss) with 25 publications. The third source is ACM international conference proceeding series which has 24 publications. The fourth source is PLOS one as well as scientific reports (19 publications) which are followed by sustainability (Switzerland) (17 publications), science of the total environment (16 publications), international archives of the photogrammetry, remote sensing and spatial information sciences—isprs archives (13 publications), 2020 international conference on AI, big data, computing and data communication systems, icabcd 2020—proceedings as well as IEEE access (12 publications), remote sensing of environment (publications 11), PLOS neglected tropical diseases (10 publications) as depicted in Table 3.

Table 2 Scholarly works over time

Year	Counts	Percentage
2022	497	30.19
2021	406	24.67
2020	286	17.38
2019	161	9.78
2018	99	6.01
2017	61	3.71
2016	44	2.67
2015	34	2.06
2014	38	2.31
2013	20	1.22
Total	1646	100

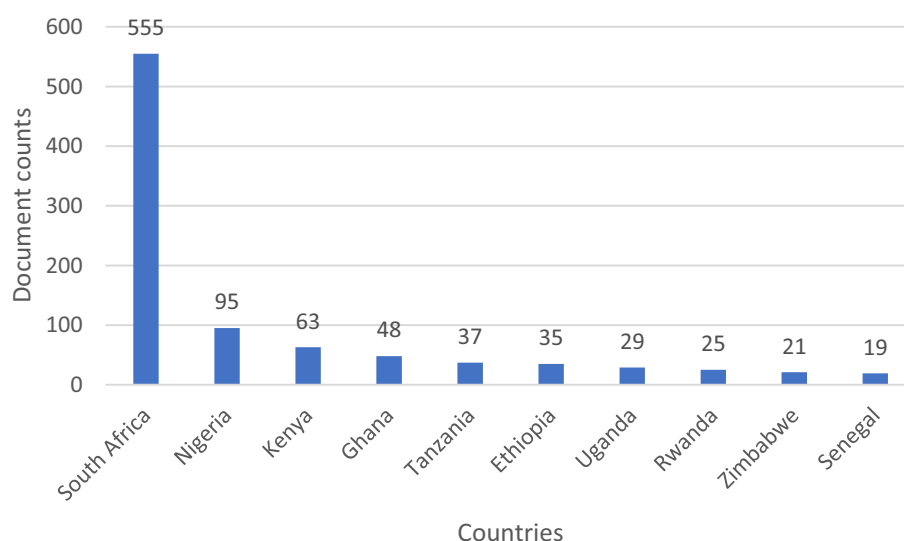
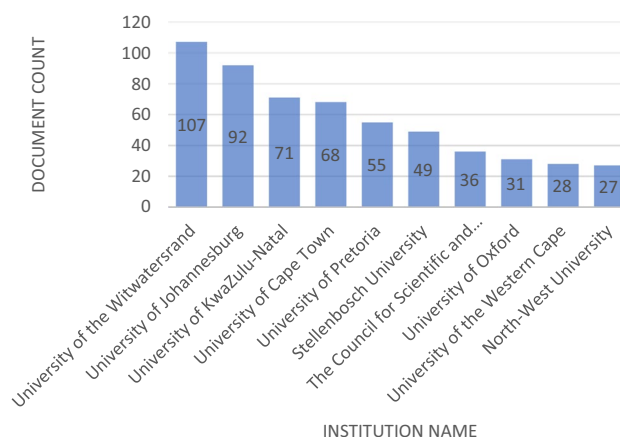
Table 3 Top sources

Source title	Document count
Remote sensing	53
International geoscience and remote sensing symposium (igarss)	25
ACM international conference proceeding series	24
PLOS one	19
Scientific reports	
Sustainability (Switzerland)	17
Science of the total environment	16
International archives of the photogrammetry, remote sensing and spatial information sciences—isprs archives	13
2020 international conference on artificial intelligence, big data, computing and data communication systems, icabcd 2020—proceedings	12
IEEE access	
Remote sensing of environment	11
PLOS neglected tropical diseases	10

**Fig. 2** Most active countries in the World

3.4 Worldwide publication trends by country

Figure 2 stresses the contributions of various nations to the development of AI research by looking at the top 10 countries in terms of publications. The results are impacted by the dataset's restriction to English-language publications. South Africa is the leading author in this field of study, contributing the most documents (555 articles). It is followed by the United States, which has 377 published articles overall. The United Kingdom comes in the third place by publishing 184 articles. Germany is in the fourth position with 113 publications. China is in the fifth place with 106 publications,

Fig. 3 Top 10 African countries**Fig. 4** Most active institutions

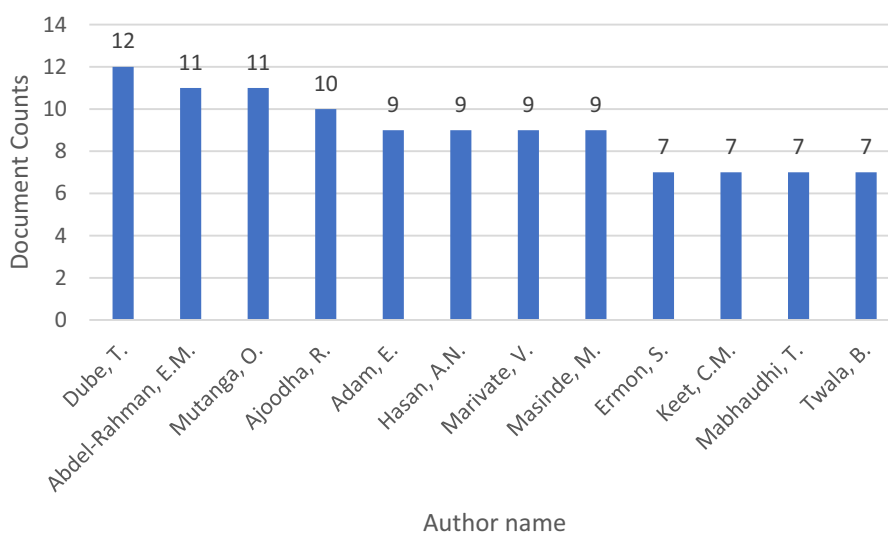
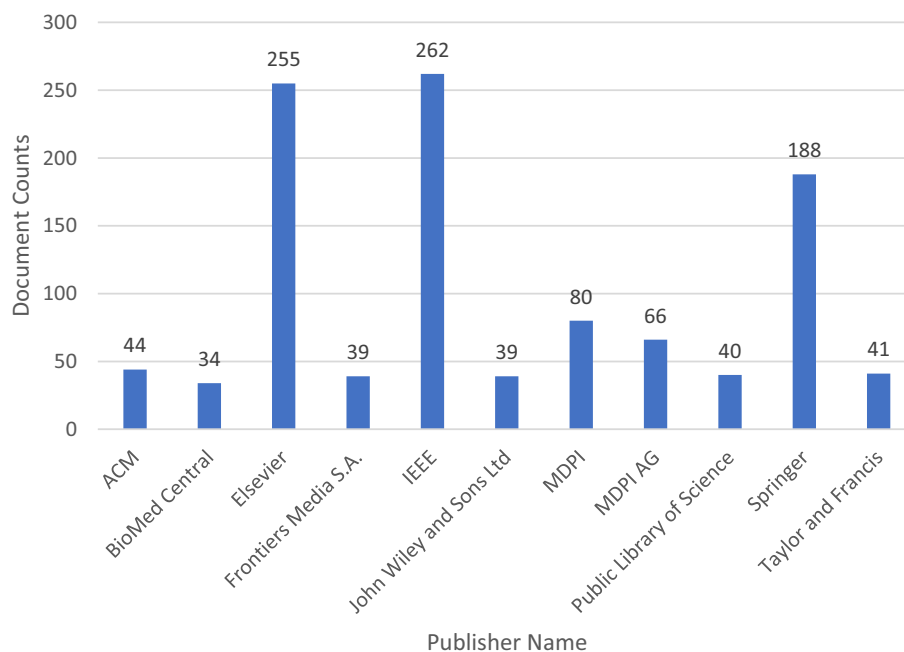
followed by Nigeria with 95 articles published, India with 76, Canada with 73, France with 67, and the Netherlands with 65. It's noteworthy to note that, in the world rankings of countries for their contributions to the expansion of AI research in Africa, South Africa tops the list, with Nigeria coming in at number six.

3.5 Productivity by country in Africa

The research identifies the African nations where articles in AI are published, as shown in Fig. 3. The country with the most papers published in AI was South Africa, with 555 publications, followed by Nigeria (95 publications), Kenya (63 publications), Ghana (48 publications), Tanzania (37 publications), Ethiopia (35 publications), Uganda (29 publications), Rwanda (25 publications), Zimbabwe (21 publications), and Senegal (19 publications).

3.6 Most active institutions

Results in Fig. 4 indicate that University of the Witwatersrand is at the top of the most active institutions in the world concerning the research in AI. The University of Johannesburg is the second. The University of KwaZulu-Natal is the third while the University of Cape Town is the fourth, followed by the University of Pretoria, Stellenbosch University, the Council for Scientific and Industrial Research, University of Oxford, University of the Western Cape and North-West University. It should be noted that the first five institutions in the list of the top ten active institutions are from South Africa. This implies that, South Africa institutions are making more efforts to publish research articles in the field of AI. Therefore, other African institutions should pull up their socks to improve the world's current status in AI research.

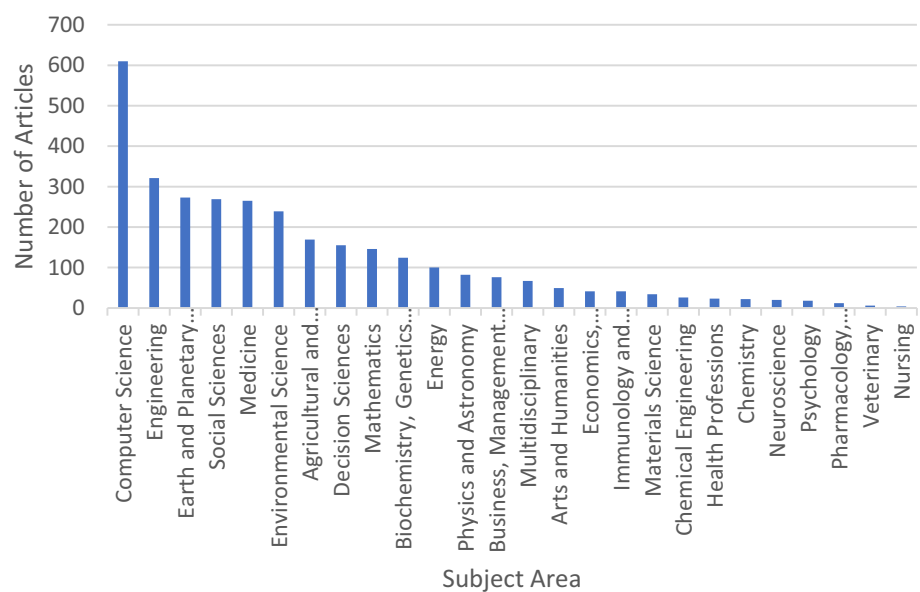
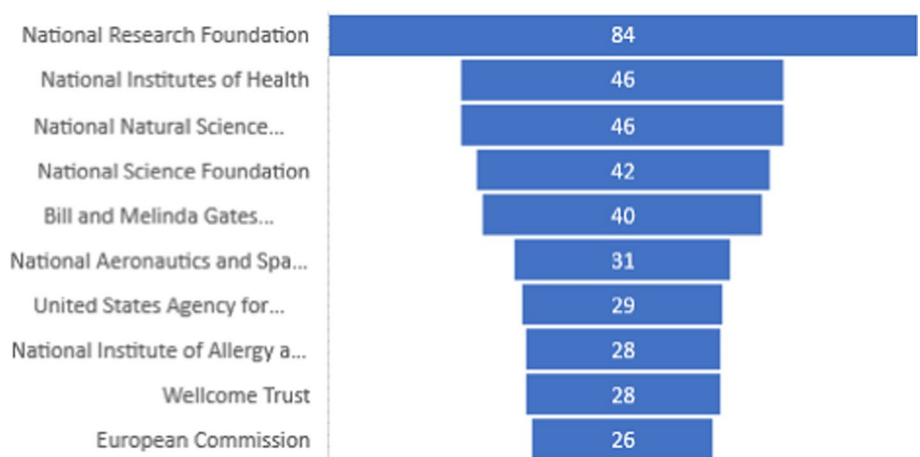
Fig. 5 Most active authors**Fig. 6** Top publishers

3.7 Records by authorship

The results presented in Fig. 5 show that, Dube, T. ranked first in the list of authors with 12 publications, followed by Abdel-rahman, E. M. with 11 publications, Mutanga, O. (11 publications), Ajoodha, R. (10 publications), Adam, E. (9 publications), Hasan, A. N. (9 publications), Marivate, B. (9 publications), Masinde, M. (9 publications), Ermon, S. (7 publications), Keet, C. M. (7 publications), Mabhaudhi, T. (7 publications), and Twala, B. (7 publications).

3.8 Records by publisher

The findings in Fig. 6 indicate the publishers where AI papers were published. Institute of Electrical and Electronics Engineers (IEEE), with 262 publications, was the first, followed by Elsevier (255 publications), Springer (188 publications), MDPI (80 publications), MDPI AG (66 publications), Association for Computing Machinery (ACM) with 44 publications,

Fig. 7 Top Subject Areas**Fig. 8** Funding bodies

Taylor and Francis (41 publications), Public Library of Science (40 publications), John Wiley and Sons Ltd (39 publications), Frontiers Media S.A. (39 publications), and BioMed Central (34 publications).

3.9 Subject area

As shown in Fig. 7, which demonstrates that the vast majority of articles published in the computer science were analyzed with a total 610 articles, followed by engineering with 321 articles, Earth and Planetary Sciences (273 publications), Social Sciences (269 publications), Medicine (265 publications), Environmental Science (239 publications), Agricultural and Biological Sciences (169 publications), Decision Sciences (155 publications), Mathematics (146 publications), Biochemistry, Genetics and Molecular Biology (124 publications), and Energy (100 publications).

3.10 Funding bodies

Research activities are rigorous, making them challenging to engage in without the proper infrastructure and financing. This section provides information on the funding organizations both inside and outside of Africa that made it possible for researchers working in the field of AI in Africa to publish their findings. According to the data in Fig. 8, the National Research Foundation topped the list of institutions that funded research projects that led to publications on artificial intelligence in Africa, followed by the National Institutes of Health, the National Natural Science Foundation of China, the

National Science Foundation, the Bill and Melinda Gates Foundation, the National Aeronautics and Space Administration, the United States Agency for International Development, and the National Institute of Allergy and Infectious Diseases.

3.11 Analysis of co-authorship of countries

To determine the network of cooperating countries, the researchers looked at national co-authorship. After establishing 30 as the minimum number of country documents and unticking the box to exclude documents coauthored by many countries, 19 of the 153 countries achieved the benchmarks. Figure 9 shows how the connections between the 19 countries were established. This shows that the 19 countries have a close relationship and have collaborated to make a significant contribution to the field. Also noteworthy is the fact that the study's normalization strategy was association. Both the minimum cluster size and the clustering resolution were set to one.

In essence, articles on AI were published in 19 different nations. The cooperative relationships between the nations or regions are shown in Fig. 9. Notably, the USA had the most interactions with other nations, with 6010 citations (18 links and 395 total link strength), followed by South Africa (4372 citations, 18 links and 277 total link strength), United Kingdom (3212 citations, 18 links and 275 total link strength) as indicated in Table 4. Furthermore, among the most cited countries, three are in cluster 1, two are in cluster 2, four are in cluster 3, and one is in cluster 4.

3.12 Keyword co-occurrence

The conceptual foundations of a domain may be made clearer by a keyword analysis [10]. According to [11], keywords highlight authors' views on how concepts and studies are grouped and related. To assess whether two phrases are related, their distance from one another is measured. The closer two terms are to one another, the more related and frequently

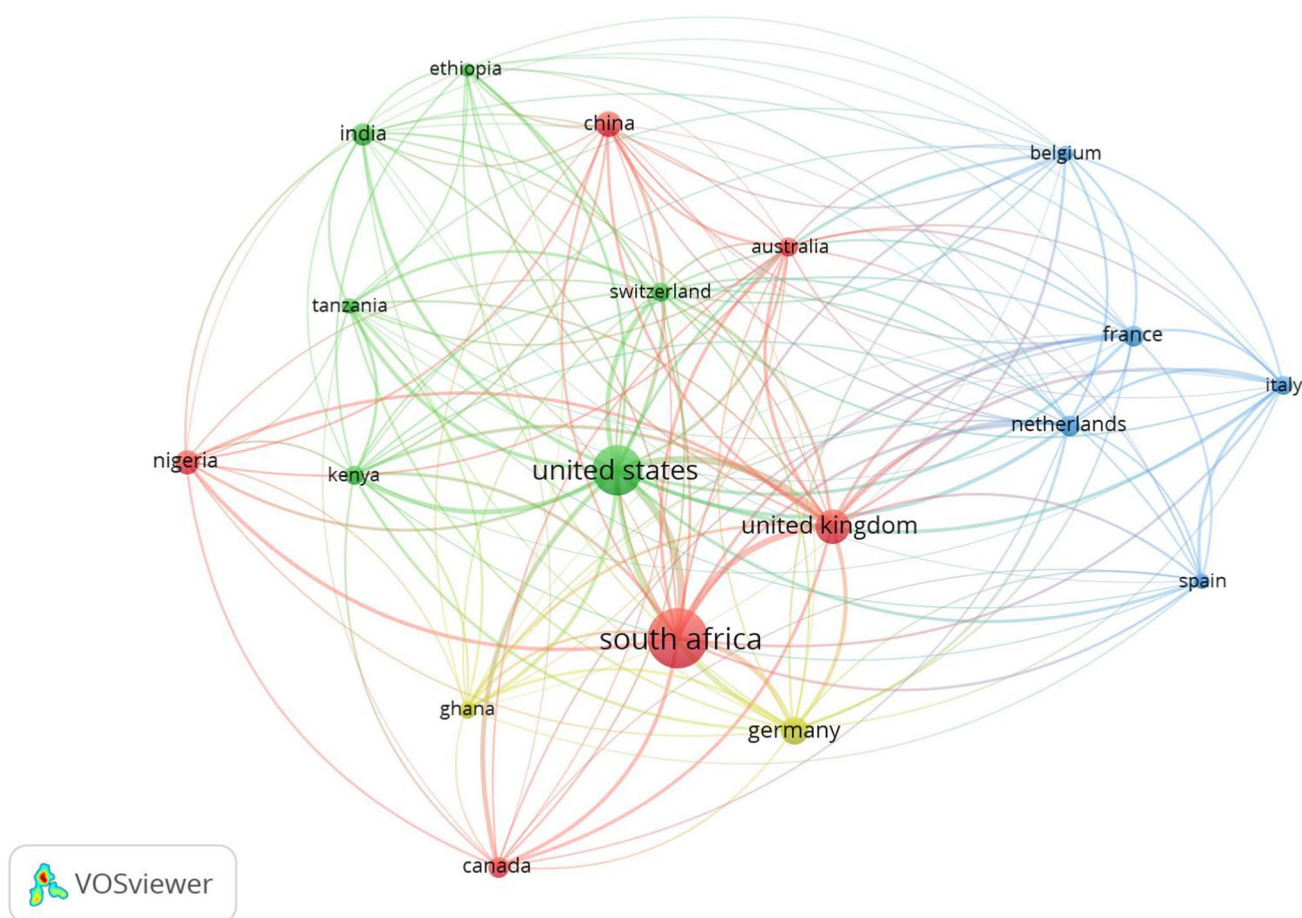


Fig. 9 Network visualization diagram based on Co-authorship of countries

Table 4 The most cited countries

Country	Documents	Citations	Cluster	Links	Total link strength
United States	377	6010	2	18	395
South Africa	555	4372	1	18	277
United Kingdom	184	3212	1	18	275
Germany	113	1857	4	18	150
Italy	57	1810	3	15	85
France	67	1277	3	16	93
Netherlands	65	1157	3	17	110
Belgium	43	1139	3	15	71
India	76	1131	2	16	57
China	106	909	1	16	79

they co-occur. The power of a word's overall length influences how frequently it appears in a manuscript. We set the minimum keyword occurrence at 20, and 17 keywords satisfied the cut-off in order to offer the term with the highest total link strength (TLS). Association strength was used as a normalization method [12]. Both the minimum cluster size and the clustering resolution were set to one. Four clusters with 84 links and a total of 424 link strengths were discovered. Clusters 1, 2, 3, and 4 each have seven, five, three, and two keywords, respectively. Several terms related to cluster 1 include deep learning, neural networks, and machine learning. It is crucial to notice that among the top 10 author keywords provided in Table 5, machine learning, artificial intelligence, and deep learning are the most common and pertinent terms. It is significant that cluster 1 contains deep learning and machine learning, and cluster 2 has artificial intelligence.

VOSviewer offers three different graphics to demonstrate bibliometric mapping as shown in Figs. 10, 11, and 12. The labels for the keywords were colored circles. The number of times a keyword appears in an abstract or title is inversely related to the size of the circle. Because of this, the frequency of letters and circles determined their size. The keyword appears more frequently as the letters and circles become more significant.

4 Discussions

This study provides a comprehensive overview of the evolution and current state of AI research in Africa. The remarkable surge in AI-related publications during this period reflects a continent-wide recognition of AI's potential to drive innovation, economic growth, and social progress. This heightened interest signals a departure from traditional resource-intensive industries to knowledge-based sectors, marking a significant paradigm shift in Africa's development trajectory. One notable trend identified in the analysis is the emergence of collaborative networks among African nations and international partners. These collaborations signify a growing recognition of the importance of knowledge exchange and capacity building in AI. By pooling resources and expertise, African researchers are fostering

Table 5 The most common author keywords

Keyword	Occurrences	Cluster	Links	Total link strength
Machine learning	375	1	16	221
Artificial intelligence	142	2	12	97
Deep learning	125	1	14	77
Africa	95	2	14	76
Remote sensing	60	1	11	54
Covid-19	48	2	9	49
Random forest	61	3	10	48
South Africa	60	3	12	44
Big data	29	2	9	31
Prediction	25	1	10	27

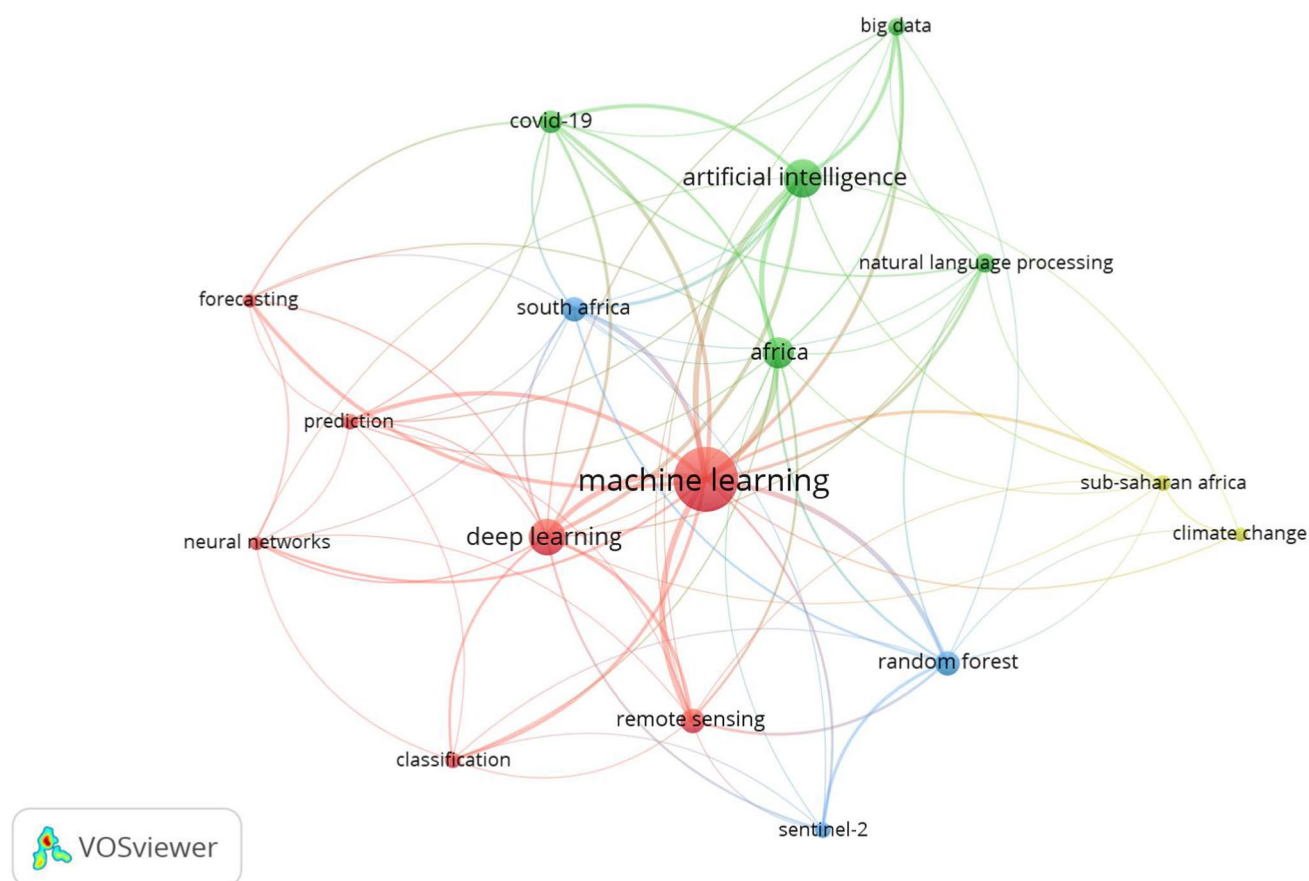


Fig. 10 Network visualization diagram based on co-occurrence of authors' keywords

a vibrant AI research community that can address region-specific challenges effectively. This collaborative spirit is not only crucial for knowledge dissemination but also for bolstering Africa's global presence in the AI research landscape.

The bibliometric analysis highlights key research themes that resonate with Africa's unique socio-economic landscape. AI applications in healthcare, agriculture, finance, and education take center stage, mirroring the continent's pressing needs. For instance, AI can revolutionize healthcare delivery in remote areas, optimize agricultural practices to ensure food security, enhance financial inclusion, and provide innovative educational solutions to bridge the learning divide. The focus on these themes underscores AI's potential to address some of Africa's most daunting challenges, thereby improving the quality of life for millions of people. Moreover, the identification of leading contributors and funding sources in AI research offers valuable insights for shaping future strategies. By recognizing the institutions and individuals driving AI research in Africa, stakeholders can foster collaboration, mentorship, and knowledge transfer to nurture the next generation of AI experts. Additionally, understanding the funding landscape helps policymakers and researchers navigate resource allocation, ensuring sustained growth in AI research and development.

The list of authors in the AI research domain in Africa reveals some prolific contributors who have made substantial contributions to the field. T. Dube takes the lead with an impressive 12 publications, indicating a strong dedication to advancing AI knowledge. Right behind Dube, E. M. Abdel-rahman and O. Mutanga share the second position with 11 publications each, showcasing their consistent engagement in AI research. Further down the list, R. Ajoodha, E. Adam, A. N. Hasan, B. Marivate, M. Masinde, S. Ermon, C. M. Keet, T. Mabhaudhi, and B. Twala are notable authors, each with significant contributions ranging from 7 to 10 publications. Their work signifies a diverse range of AI interests and expertise, reflecting the multifaceted nature of AI applications in Africa. These authors not only contribute to the growth of AI research in Africa but also serve as potential mentors and collaborators, shaping the future of AI development on the continent. Their collective efforts play a pivotal role in addressing Africa's unique challenges and opportunities through the application of artificial intelligence.

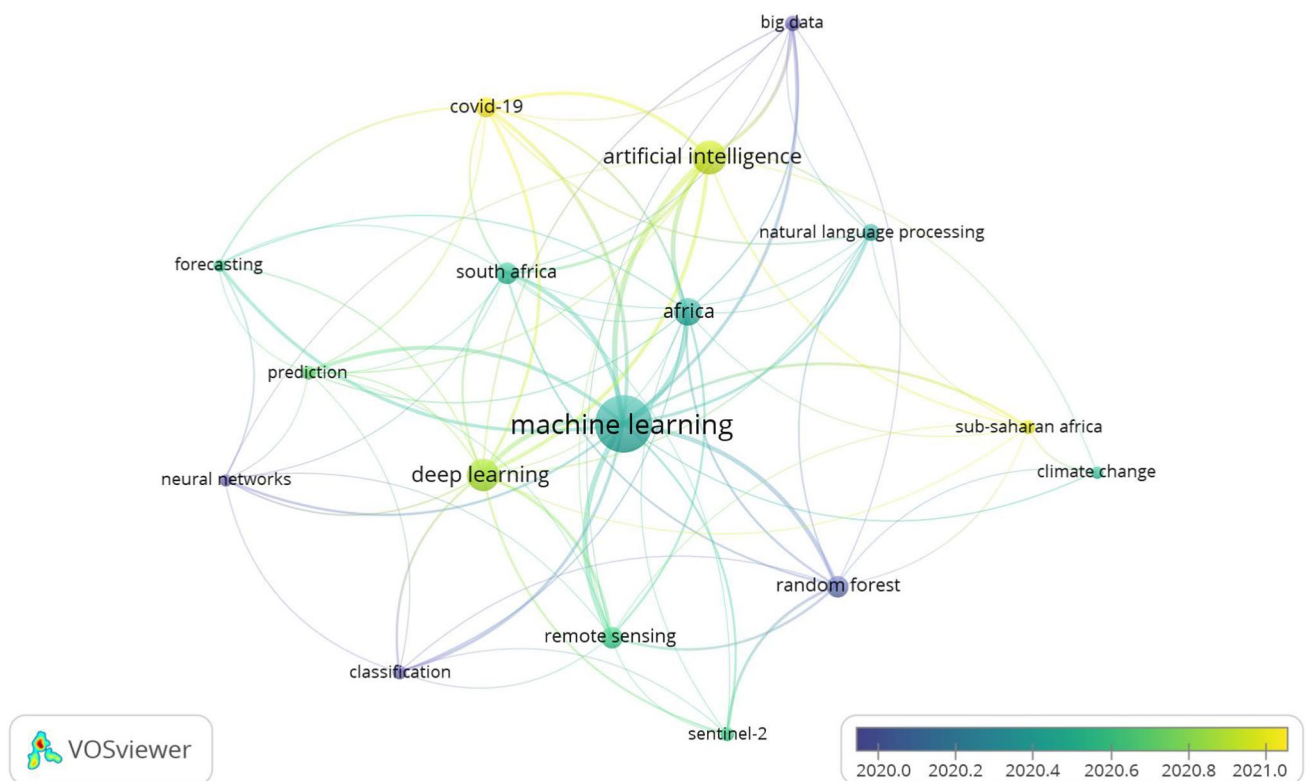


Fig. 11 Overlay visualization

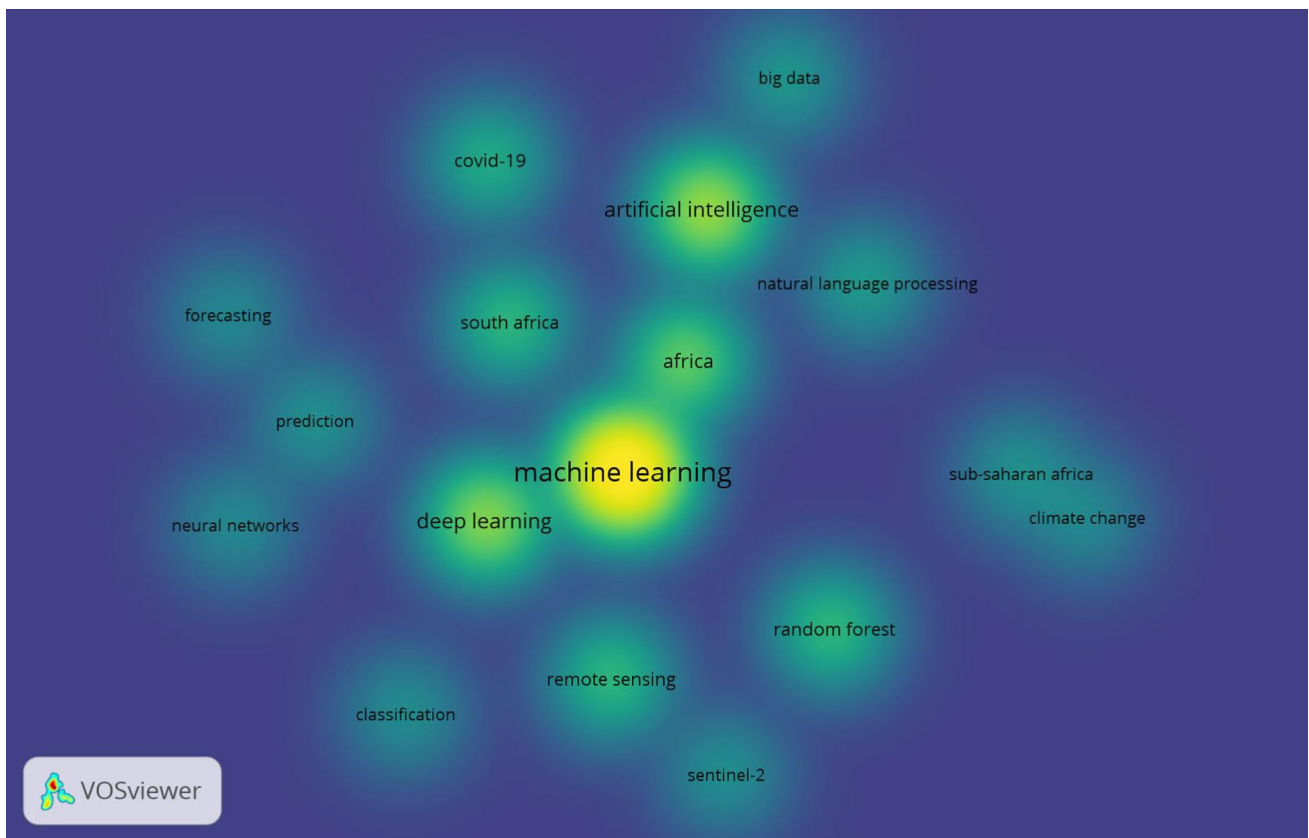


Fig. 12 Density visualization

Despite the valuable insights gained in this study, two limitations need to be acknowledged. Firstly, the analysis primarily relies on data available in the Scopus database, which may not encompass all AI-related publications in Africa. This database may not include research published offline in local or regional journals that do not have international visibility, potentially leading to an incomplete representation of AI research efforts on the continent. Secondly, the exclusion of non-English language articles could introduce language bias, as valuable research in other languages might exist, particularly in North and Francophone Africa. Future research can address issues with the research methodology related to coverage and language bias.

5 Conclusions and recommendations

5.1 Conclusions

The discrepancy in research output among African countries emphasizes the need for more funding in the areas that use it the most. The researchers looked into every piece of AI research published in Africa on April 2, 2023. Therefore, the most common form of material are research papers that were published in journals (952 documents), followed by articles from conference proceedings (466 publications), and reviews (70 publications). The source with the most significant number of publications is Remote Sensing with 53 papers, followed by international geoscience and remote sensing symposium (igarss) with 25 publications and ACM international conference proceeding series which has 24 publications. Additionally, it was found that South Africa, the United States, and the United Kingdom had published many papers, with their total worth accounting for more than half of all publications.

Moreover, University of the Witwatersrand is the most active institution, with the maximum number of publications. The journal, Remote Sensing, ranked at the top regarding publishing the AI related scholarly works. Dube, T. was at the top of the list of published authors with 12 publications. Both Abdel-rahman, E. M. and Mutanga, O. followed him with 11 publications. Furthermore, IEEE with 262 publications is the first on the list of top publishers in the world. Furthermore, machine learning, followed by artificial intelligence and deep learning, are the most frequent and relevant keywords among the 10 most popular author keywords. The United States was the most cited country followed by South Africa and the United Kingdom. In addition, the National Research Foundation ranked first in the list of funders that supported research activities on AI in Africa, followed by the National Institutes of Health, and National Natural Science Foundation of China.

5.2 Recommendations

Based on the insights derived from this study, several strategic recommendations can be made to advance the state of AI research and its practical application in Africa. Firstly, it is imperative to promote and invest in interdisciplinary collaborations. Encouraging researchers from diverse fields, such as computer science, healthcare, agriculture, economics, and social sciences, to work together can lead to holistic AI solutions that address complex challenges more effectively. Funding agencies and institutions should facilitate such cross-disciplinary partnerships and support research endeavors that bridge the gap between AI theory and real-world problem-solving. Secondly, fostering an ecosystem for innovation and entrepreneurship in AI is essential. African governments, in collaboration with industry leaders and academia, should establish innovation hubs, incubators, and startup accelerators specifically focused on AI. These initiatives can provide resources, mentorship, and funding to nurture AI startups and encourage the development of AI-driven solutions tailored to Africa's unique needs.

Furthermore, AI education and training programs should be expanded at both the undergraduate and postgraduate levels to cultivate a skilled AI workforce capable of driving innovation and technological adoption across sectors. In parallel, efforts should be made to create clear ethical and regulatory frameworks that guide AI development and ensure its responsible use, addressing concerns related to bias, transparency, and data privacy. By implementing these recommendations, Africa can harness the full potential of artificial intelligence, fostering sustainable development and enhancing its global presence in the AI research community.

Author contributions "T.K. had the idea for the article and performed the literature search and data analysis. In addition, T.K. wrote the main manuscript text. S.D. critically revised the work. All authors reviewed the manuscript."

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Data availability Not applicable.

Code availability Not applicable.

Declarations

Competing interests On behalf of all authors, the corresponding author states that there is no conflict of interest.

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References

1. Tomašev N, Cornebise J, Hutter F, Mohamed S, Picciariello A, Connelly B, et al. AI for social good: unlocking the opportunity for positive impact. *Nat Commun*. 2020;11(1):1–6.
2. Górriz JM, Ramírez J, Ortíz A, Martínez-Murcia FJ, Segovia F, Suckling J, Leming M, Zhang YD, Álvarez-Sánchez JR, Bologna G, Bonomini P, Casado FE, Charle D, Charle F, Contreras R, Cuesta-Infante A, Duro RJ, Fernández-Caballero A, Fernández-Jover E, et al. Artificial intelligence within the interplay between natural and artificial computation: advances in data science, trends and applications. *Neurocomputing*. 2020;410:237–70. <https://doi.org/10.1016/j.neucom.2020.05.078>.
3. Thomas N. AI in radiology and pathology: a brief introduction and ethical considerations. *J India Dent Assoc Kochi*. 2021;3(1):12–7.
4. Dobrev D. A definition of artificial intelligence. *arXiv preprint*; 2012. [arXiv:1210.1568](https://arxiv.org/abs/1210.1568). <https://doi.org/10.48550/arXiv.1210.1568>.
5. Zhang C, Lu Y. Study on artificial intelligence: the state of the art and future prospects. *J Ind Inf Integr*. 2021;23: 100224. <https://doi.org/10.1016/j.jii.2021.100224>.
6. Samuel AL. Some studies in machine learning using the game of checkers. II-recent progress. In: *Computer games*. New York: Springer; 1988. p. 366–400.
7. Tang A, Tam R, Cadrin-Chenevert A, Guest W, Chong J, Barfett J, Chepelev L, Cairns R, Mitchell JR, Cicero MD, Poudrette MG, Jaremko JL, Reinhold C, Gallix B, Gray B, Geis R. CAR white paper on AI in radiology. *Can Assoc Radiol J*. 2018;69:120–35.
8. LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature*. 2015;521(7553):436.
9. Hirschberg J, Manning CD. Advances in natural language processing. *Science*. 2015;349:261–6. <https://doi.org/10.1126/science.aaa8685>.
10. Zhang D, Zhang D, Managi S. A bibliometric analysis on green finance: Current status, development, and future directions. *Fin Res Lett*. 2019;29:425–30. <https://doi.org/10.1016/j.frl.2019.02.003>.
11. Danvila-del-Valle I, Estévez-Mendoza C, Lara FJ. Human resources training: a bibliometric analysis. *J Bus Res*. 2019;101:627–36. <https://doi.org/10.1016/j.jbusres.2019.02.026>.
12. Saheb T, Saheb T, Carpenter DO. Mapping research strands of ethics of artificial intelligence in healthcare: a bibliometric and content analysis. *Comput Biol Med*. 2021;135: 104660.

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